MATH 137 Fall 2020: Practice Midterm 1

Multiple Choice

MC01. For the sequence $\{a_n\}$ where $a_1 = 1$, $a_n = \sqrt{6 + a_{n-1}}$ for $n \geq 2$. The value of $\lim_{n\to\infty} a_n$ is

(a) -3
(b) -2
(c) 2
(d) None of the above. [redacted]

MC02. $\lim\limits_{x\to 1} \dfrac{\sin(x-1)}{x^2-1}$

(a) $= 1$.
(b) $= \frac{1}{2}$. [redacted]
(c) Does not exist.
(d) None of the above.

MC03. If f is not continuous at $x = 2$ then it must be the case that

(a) $f(2) \geq 0$.
(b) $f(2)$ is undefined.
(c) $f(2)$ is defined.
(d) None of the above. [redacted]

MC04. The sequence defined by $a_n = \dfrac{n^2+1}{n+3}$

(a) converges.
(b) is non-increasing.
(c) is bounded below. [redacted]
(d) None of the above.

MC05. If $f(x) = 7$ for all $x \in \mathbb{R}$ then $f'(x)$

(a) exists for all $x \in \mathbb{R}$. [redacted]
(b) is not continuous for all $x \in \mathbb{R}$.
(c) $= 1$.
(d) None of the above.

True/False

TF06. Three functions, f, g and h, are defined on an open interval I containing $x = a$. If for each $x \in I$, $g(x) < f(x) < h(x)$ and $\lim\limits_{x\to a^+} g(x) = L = \lim\limits_{x\to a^+} h(x)$, then $\lim\limits_{x\to a} f(x) = L$.

[redacted]

TF07. The Fundamental Trigonometric Limit tells us that if θ is small, then $\cos\theta \approx \theta$.

[redacted]

TF08. If f is continuous on $\mathbb{R}$ and $f(0) > 0$ then there exists $\delta > 0$ so that $f(x) > 0$ for all $x \in (0, \delta)$.

[redacted]

TF09. $\lim\limits_{x\to\infty} \dfrac{x^p}{e^x} = 0$ for all $p \in \mathbb{R}$.

[redacted]

TF10. If $f(a)$ exists, then $f'(a)$ exists too.

[redacted]

Short Answer

SA01. For the function

$$f(x) = \begin{cases} 1 + \sin x & x < 0 \\ \cos x & 0 \le x \le \pi \\ \sin x & \pi < x \end{cases}$$

determine

(a) $\lim\limits_{x\to 0} f(x)$, or write DNE if it does not exist.

[redacted]

(b) $\lim\limits_{x\to \pi} f(x)$, or write DNE if it does not exist.

[redacted]

SA02. Write all solutions to $|x - 1| = |2x|$

[redacted]

SA03. Give an example of a function such that the Extreme Value Theorem does not apply to it on the interval $[0, 5]$.

[redacted]

SA04. State the formal ϵ–δ definition of what it means for $\lim_{x \to a} f(x) = L$.

[redacted]

Long Answer

LA01. Use the ϵ–N definition of the limit of a sequence to show $\lim_{n \to \infty} \dfrac{n+1}{3n+2} = \dfrac{1}{3}$.

[redacted]

LA02. Compute the following sequence limits, or show that they do not exist.

(a) $\lim_{n \to \infty} \dfrac{\cos n}{n}$

[redacted]

(b) $\lim_{n\to\infty} \dfrac{2n^2 - n - 1}{5n^2 + n - 3}$

[redacted]

LA03. Consider the recursive sequence $a_1 = 5$ and $a_{n+1} = \dfrac{a_n + 1}{3}$ for $n \geq 1$.

(a) Prove that the sequence is decreasing and is bounded below by 0.

[redacted]

(b) Prove that the sequence converges and find its limit.

[redacted]

LA04. Use the ϵ–δ definition of the limit of a function to show that $\lim_{x\to 2} \left(x^2 + 2x - 3\right) = 5$.

[redacted]

LA05. Compute the following function limits, if possible. If the limit does not exist, prove it.

(a) $\lim_{x\to 0} \dfrac{5x^2 - 3x}{2x^3 - x^2}$

(b) $\lim_{x\to 0} \dfrac{3x^2 - 1}{x^2 - x + 1}$

(c) $\lim_{x\to \infty} \dfrac{\ln x^2 - \ln x}{x^2 - x}$

LA06. Given the function

$$f(x) = \begin{cases} k^2 x + 5 & x \leq -2 \\ x^2 + k & x > -2 \end{cases}$$

If $f(x)$ is continuous at $x = -2$, determine all value(s) for k.



LA07. Determine, with justification, all vertical asymptotes of the function

$$f(x) = \frac{x+3}{|x^2 - 2x - 15|}.$$



LA08. Suppose $A, B \in \mathbb{R}$, $A > 0$, $B > 0$ and $f : \mathbb{R} \to \mathbb{R}$ is a function such that if $|x - y| < A$ then $|f(x) - f(y)| < B|x - y|$ for all $x, y \in \mathbb{R}$.

Prove that f is continuous on $\mathbb{R}$.





LA09. Prove that $x^2 + x\cos x = 1$ has at least two real solutions.

